INSTITUTIONAL Portfolio Optimizer

[Link of Github](#)

STRATEGIC Portfolio GENERATION

Advanced Mathematical Modeling for Institutional Portfolios

EXECUTIVE SUMMARY & PORTFOLIO SCORECARD

INVESTMENT PHILOSOPHY Risk is allocated first. Capital is allocated second. Objective: maximize risk-adjusted returns within predefined risk budgets.

13.37%
Return Forecast
Annual Expected

+6.54%
Expected Alpha
vs Benchmark

13.30%
Portfolio Volatility
Annual Std Dev

0.830
Portfolio Beta
Market Sensitivity

-14.31%
Max Drawdown
Peak-to-Trough

0.694
Fund. Score
Weighted Average

TARGET VS ACHIEVED SCORECARD

Metric	Target	Achieved	Status
Return Forecast	12–14%	13.37%	✅
Alpha	6–7%	6.54%	✅
Volatility	<14%	13.30%	✅
Drawdown	>-15%	-14.31%	✅
Beta	0.85–0.95	0.830	⚠️
Sharpe Ratio	>0.65	0.516	⚠️

KEY MESSAGE A successful portfolio is not one that maximizes every metric. It is one that remains inside its risk budgets while delivering attractive expected returns.

FRAMEWORK METHODOLOGY: MODELS, ASSUMPTIONS & LIMITATIONS

EXECUTIVE TAKEAWAY Institutional transparency requires acknowledging not just how the framework works, but its foundational assumptions and boundary conditions.

MATHEMATICAL MODELS ENGINE

- **Primary Solver:** SLSQP
- **Fallback Solver:** Trust-Constr
- **Penalty System:** L2 Regularization
- **Risk Matrix:** Covariance

CORE ASSUMPTIONS

A1 Historical Stationarity
Historical covariance & standard deviation are reliable proxies for future market volatility.

A2 Fundamental Score Validity
Pre-filtration scores (ROE, Margin, ROCE) accurately eliminate structurally weak businesses.

A3 Market Efficiency (Partial)
Prices reflect publicly available information; alpha is earned through rigorous quality screening.

A4 Constraint Enforceability
Portfolio constraints (position limits, sector caps) can be maintained through rebalancing.

⚠ SHORTCOMINGS & LIMITATIONS

RISK Backward-Looking Risk
VaR & covariance matrices may fail to predict unprecedented 'Black Swan' events, historical data does not capture regime shifts.

COST Frictionless Allocation
Optimization does not model liquidity slippage or transaction costs, live implementation will face execution drag.

DATA Fundamental Score Lag
Financial statement data has quarterly delays; scores may not reflect very recent deterioration in business quality.

BIAS Estimation Error
Expected return inputs are model-derived estimates, not realized values; optimizer can amplify small return mis-estimates.

DECK ROADMAP: PRESENTATION FLOW

01 Architecture & Filtration(Slide 4-6)
Portfolio construction layers, deep fundamental pre-filtration & quality threshold

→

02 Constraints & Optimization(Slide 7)
SLSQP / Trust-Constr engines, diversification rules, concentration & weight limits

→

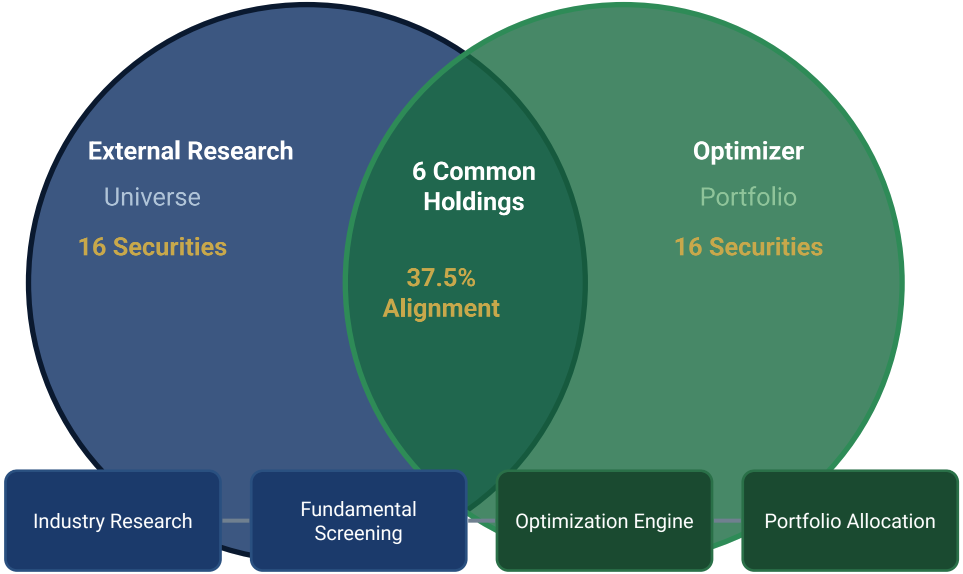
03 Risk Budgeting & Preservation(Slide 8-9)
95% volatility & drawdown budget utilization, Beta trade-off, capital preservation mandate

→

04 Final Allocation & Pipeline(Slide 10-11)
16-security target portfolio, investment lifecycle, technology stack & scalability

TRANSLATING SECURITY RESEARCH INTO PORTFOLIO CONSTRUCTION

EXECUTIVE TAKEAWAY Investment ideas are inputs. Portfolio construction creates the final investment product.



Advisory Universe

16 securities from external research

Optimizer Portfolio

16 securities via mathematical optimization

Common Holdings with Top Portfolio Advisories

6 securities — validated by both approaches

Independent Additions

10 optimizer-only positions for diversification

Research generates ideas. Portfolio construction determines how those ideas become investable, risk-controlled allocations.

PORTFOLIO CONSTRUCTION ARCHITECTURE

EXECUTIVE TAKEAWAY Every security serves a predefined portfolio role. The portfolio is diversified by function, not merely by security count.

Core Compounders (~36.5%)
Role: Volatility Anchor
Marico · Nestlé · Sun Pharma · ICICI Bank

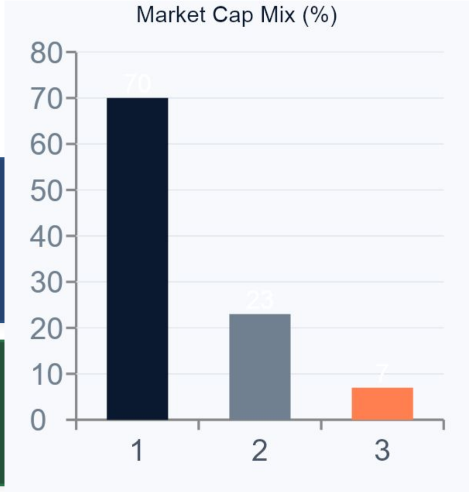
Alpha Engines (~56.5%)
Role: Primary Return Generation
Bharti Airtel · Atul · UltraTech · Choice Intl

Tactical Satellites (~7%)
Role: Controlled Upside
JSW Steel

TACTICAL SATELLITES
~7% | Controlled Upside

ALPHA ENGINES
~56.5% | Primary Return Generation

CORE COMPOUNDERS
~36.5% | Volatility Anchor & Fundamental Quality



Portfolio architecture allocates capital to layers by function — anchoring volatility with defensive quality, driving returns through high-conviction growth, and capping risk through satellite position limits.

DEEP FUNDAMENTAL PRE-FILTRATION

EXECUTIVE TAKEAWAY Optimization begins only after quality control. The optimizer is never allowed to allocate capital to structurally weak businesses.

Broad Universe
500+ Stocks

Industry Screen
~120 Candidates

Fundamental Quality Filter
~40 Qualified

Portfolio Holdings
16 Securities

Portfolio Weighted Avg Fundamental Score **0.694**

QUALITY FILTRATION METRICS

- **Operating Margin** Profitability efficiency at operational level
- **Revenue Growth** Sustainable top-line expansion trajectory
- **Debt-to-Equity** Balance sheet strength and leverage control
- **Return on Equity (ROE)** Shareholder value creation efficiency
- **ROCE** Capital deployment effectiveness
- **Industry Relative Quality** Peer benchmarking for sector context

QUALITY THRESHOLD Fundamental Score < 0.45 → Automatically Eliminated

INSTITUTIONAL PORTFOLIO CONSTRUCTION FRAMEWORK

EXECUTIVE TAKEAWAY The optimizer allocates risk before allocating capital. Mathematical constraints enforce discipline that humans cannot maintain emotionally.

OPTIMIZATION ARCHITECTURE

Expected Return Vector

Risk Constraints Applied

SLSQP Primary Optimizer

Trust-Constr Fallback

Final Capital Allocation

PORTFOLIO CONSTRAINTS

Industry Diversification
Max 2 securities per industry

Concentration Limit
11.5% maximum allocation ceiling

Minimum Meaningful Weight
2.0% minimum allocation floor

Volatility Budget
Volatility target below 14.0%

Drawdown Budget
Max drawdown target above −15%

SLSQP Primary solver — Sequential Least Squares Programming. **Trust-Constr** Fallback for complex constraint convergence. Both engines ensure feasible, risk-bounded allocations.

RISK BUDGETING FRAMEWORK

EXECUTIVE TAKEAWAY Portfolio risk is treated as a finite resource that must be allocated intentionally — not a residual outcome of security selection.

RISK BUDGET TARGETS

Risk Metric	Budget	Achieved	Utilization
Volatility	< 14.0%	13.30%	95%
Max Drawdown	> −15.0%	−14.31%	95%
Portfolio Beta	0.85–0.95	0.830	Trade-off
Concentration	Max 11.5%	Controlled	✅
Industry Exp.	Max 2/sector	Achieved	✅

BUDGET UTILIZATION

Volatility Budget

13.30% / 14.00% → 95% utilized

Drawdown Budget

−14.31% / −15.00% → 95% utilized

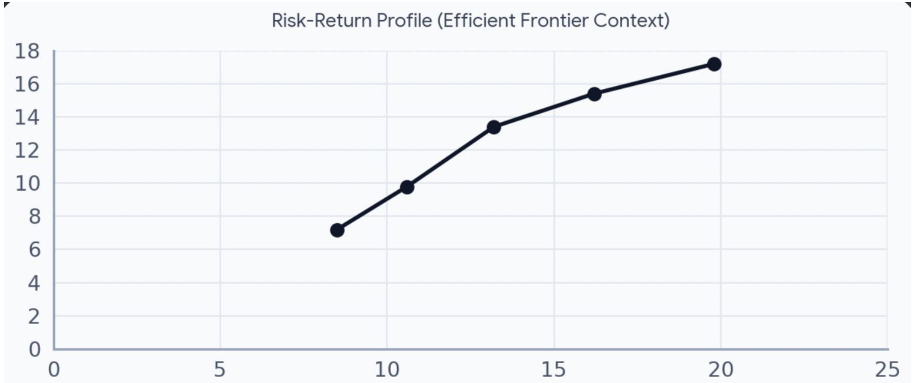
KEY INTERPRETATION
The optimizer intentionally utilized 95% of both risk budgets — extracting maximum return within defined boundaries without breaching any limit.

Risk budgeting converts qualitative investment policy statements into quantitative constraints. The optimizer treats risk as a scarce resource — allocating it deliberately, not accidentally.

CAPITAL PRESERVATION AS A FIRST PRINCIPLE

EXECUTIVE TAKEAWAY Avoiding severe capital impairment is prioritized over maximizing market exposure. Lower Beta was a deliberate outcome of the capital preservation mandate.

−14.31%	−1.34%	13.30%
Max Drawdown	**Daily VaR (95%) [A7]**	**Portfolio Volatility**



THE VOLATILITY PARADOX

Beta
Target: 0.85–0.95
Achieved: 0.830
⚠️ Trade-off

Sharpe Ratio
Target: > 0.65
Achieved: 0.516
⚠️ Trade-off

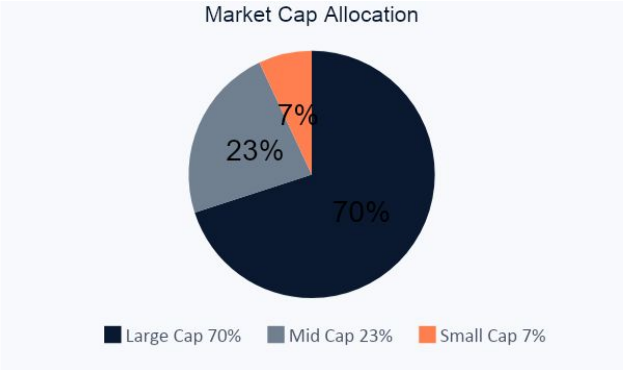
Simulation testing showed Beta >0.90 and Sharpe >0.65 required Volatility >15% and drawdowns beyond acceptable limits. The optimizer shifted capital to defensive FMCG and Healthcare.

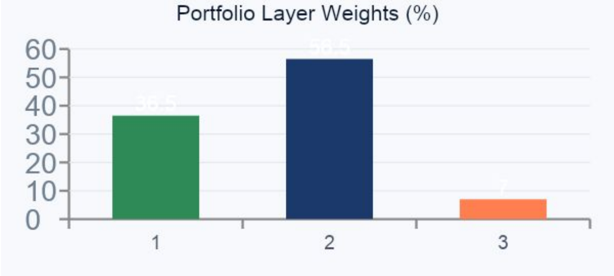
The Beta and Sharpe shortfalls are not failures — they are the cost of maintaining drawdown discipline. The optimizer correctly determined that capital preservation outweighed exposure maximization.

FINAL TARGET-LOCKED PORTFOLIO

EXECUTIVE TAKEAWAY Every allocation reflects a balance between conviction, diversification, and risk control. No position is held arbitrarily.

Security	Alloc.	Layer	Sector	Beta	F.Score
Bharti Airtel	11.5%	Alpha Engine	Telecom	1.05	0.74
Atul Ltd	10.2%	Alpha Engine	Chemicals	0.88	0.71
UltraTech Cement	9.8%	Alpha Engine	Materials	1.12	0.69
Choice Intl	8.9%	Alpha Engine	NBFC	1.21	0.65
Eicher Motors	7.4%	Alpha Engine	Auto	0.95	0.72
Marico	7.2%	Core Compounder	FMCG	0.65	0.75
Sun Pharma	6.8%	Core Compounder	Pharma	0.72	0.78
JSW Steel	7.0%	Tactical Satellite	Metals	1.35	0.62
ICICI Bank	6.5%	Core Compounder	Banking	1.08	0.73
AIA Engineering	5.5%	Alpha Engine	Industrials	0.82	0.68
3M India	5.2%	Alpha Engine	Diversified	0.76	0.70
Bosch	4.8%	Alpha Engine	Auto Anc.	0.85	0.71
Lupin	3.9%	Core Compounder	Pharma	0.68	0.71
Nestlé India	3.6%	Core Compounder	FMCG	0.55	0.72
Power Grid	3.3%	Core Compounder	Utilities	0.60	0.76
Pidilite	2.4%	Alpha Engine	Chemicals	0.90	0.69





INVESTMENT LIFECYCLE & SCALABILITY

EXECUTIVE TAKEAWAY The framework replicates a complete institutional investment workflow — from data ingestion to performance attribution.

INVESTMENT LIFECYCLE

- Research Universe
- Fundamental Screening
- Quality Scoring
- Risk Assessment
- Portfolio Optimization
- Allocation Engine
- Monitoring & Rebalancing
- Performance Attribution

TECHNOLOGY STACK

- **Python**
- **NumPy**
- **Pandas**
- **SciPy (SLSQP)**
- **yFinance**
- **Requests**

SCALABILITY FEATURES

- Dynamic data ingestion
- Automated quality scoring
- Systematic construction
- Continuous risk monitoring
- Rebalancing support
- Performance attribution

INVESTMENT COMMITTEE FINAL STATEMENT

The optimizer was designed to maximize expected returns while respecting risk budgets, diversification requirements, and capital preservation objectives. Evaluate success by risk-budget adherence, not metric maximization.

Repository

Link of Github

THANK YOU

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